4.1 Translations

Name _____ Period ____ Date _____

Geometry 4.1 (30-minute plan) · Student sheet

I can…

- name a vector and its components.
- translate a figure using a vector or a coordinate rule.
- compose translations and write one rule for them.

Vocabulary · fill in the missing word

Vector — a quantity that has both _____ and magnitude, shown as an arrow

Transformation — a _____ that moves or changes a figure in some way

Preimage and image — the original figure and the new figure after a _____

Translation — a transformation that moves every point of a figure the same distance in the same _____

Rigid motion — a _____ that preserves length and angle measure

Part 1 · Notes and examples (fill in each note, then work the example)

Vector notation: ⟨a, b⟩ means move a units horizontally and b units _____.

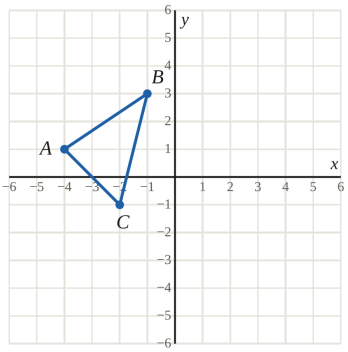
Example 1

Name the vector from A(1, 2) to B(4, −1) in component form.

Coordinate rule for a translation: (x, y) → _____

Example 2

Translate △ABC by the vector ⟨5, −2⟩. Give the image's vertices.



Composition Theorem: the composition of two (or more) rigid motions is a _____ motion.

Example 3

Translate by $(x, y) \to (x + 2, y)$, then by $(x, y) \to (x, y - 4)$. Write one rule, and find the image of $(1, 1)$.

Part 2 · Partner practice

Partner practice

A translation maps $(3, -1)$ to $(-2, 4)$. Write the rule.